AIGR TreeGrafter Function-Inference Stress Test — Final Report

Target gene: `pdxH` (*Pseudomonas putida* KT2440, PSEPK) · **UniProt:** Q88NS5 **Seed hypothesis under test:** pdxH has *pyridoxal 5'-phosphate biosynthetic process* (**GO:0042823**), propagated by TreeGrafter/PANTHER as an IEA annotation (GO_REF:0000118).

Executive Judgment

Verdict: SUPPORTED.

The TreeGrafter/PANTHER propagation of **GO:0042823 (pyridoxal 5'-phosphate biosynthetic process)** to *P. putida* pdxH (Q88NS5) is **correct**. None of the three characteristic failure modes of automated phylogenetic propagation applies:

1. **Granularity / family-vs-subfamily → NOT triggered.** Q88NS5 maps to the narrowest, characterized subfamily (PANTHER **PTHR10851:SF0**, "pyridoxine-5'-phosphate oxidase"), the *same* subfamily as the biochemically and structurally defined *E. coli* PdxH. The propagated term is at the correct granularity, not an over-broad family label.
2. **Pseudo-enzyme / loss of activity → NOT triggered.** Every critical catalytic, FMN-binding, and phosphate-binding residue is conserved relative to the characterized *E. coli* enzyme. There is no active-site degeneration.
3. **Within-superfamily mis-placement → NOT triggered.** The domain architecture (Pfam PF01243 + PF10590) and InterPro signatures place the protein unambiguously in the PNPOx family, not in a functionally distinct neighboring subfamily of the shared split β-barrel FMN-binding fold.

Single most decisive piece of evidence: Q88NS5 shares **41.5% sequence identity** over 212 aligned columns with the structurally/biochemically characterized *E. coli* PdxH (P0AFI7), falls in the **same PANTHER subfamily (PTHR10851:SF0)** with an **identical domain architecture**, and retains **all** critical catalytic/FMN/phosphate-binding residues (20/25 UniProt-annotated binding residues identical, the 5 differences all conservative substitutions).

Summary

The core question in this stress test is whether a homology-propagated GO term truly reflects the *direct* function of the query protein, or whether the graft has introduced one of three characteristic errors. For *P. putida* KT2440 pdxH (Q88NS5), the answer is clear: the propagation is **correct**. The protein is a genuine, catalytically intact **pyridoxine/pyridoxamine 5'-phosphate oxidase (PNPOx, EC 1.4.3.5)** — the enzyme that catalyzes the terminal step of de novo pyridoxal 5'-phosphate (PLP) biosynthesis and also contributes to the vitamin-B6 salvage pathway.

The confidence of this call comes from a convergence of independent evidence lines, each of which closes one of the three potential failure modes. The protein resolves to the narrow, experimentally characterized PANTHER subfamily **PTHR10851:SF0** — the same subfamily as the structurally defined *E. coli* PdxH — so the term is at the right granularity. Its two-domain architecture (**PF01243 PNPOx_N** + **PF10590 PNP_phzG_C**, with the conserved-site signature IPR019740) is identical to that reference, ruling out mis-placement onto a neighboring subfamily of the same fold. And its active site is fully preserved: 20 of 25 binding residues are identical, with every critical catalytic, FMN-, and phosphate-binding residue conserved — ruling out a pseudo-enzyme. An orthogonal, rule-based **HAMAP (MF_01629) EC 1.4.3.5** assignment independently corroborates the sequence analysis.

For curation, the recommendation is to **retain GO:0042823** and, optionally, to **add the more specific molecular-function term GO:0004733 (pyridoxamine phosphate oxidase activity)** to capture the exact catalytic activity that executes the process. No generalization, removal, or sibling-term replacement is warranted. The only meaningful knowledge gap is the absence of a *P. putida*-specific biochemical assay — all direct enzymology is from the *E. coli* (and human) orthologs — which could be closed by a straightforward PNPOx activity assay or an *E. coli ΔpdxH* complementation test.

Independent Family/Function Assignment

Independent of the propagated process term, my own analysis assigns Q88NS5 the following **specific molecular function**:

- **Molecular function:** pyridoxine/pyridoxamine 5'-phosphate oxidase activity (FMN-dependent) — **GO:0004733** (pyridoxamine phosphate oxidase activity), **EC 1.4.3.5**.

- **Characterized homolog basis:** *Escherichia coli* pyridoxine/pyridoxamine 5'-phosphate oxidase (**PdxH / PNPOx**, UniProt **P0AFI7**), whose crystal structures, FMN complex, and catalytic mechanism are published (PMID: 12686112, PMID: 10903950, PMID: 15858270).
- **GO terms I would assign:**
- **BP: GO:0042823** (pyridoxal 5'-phosphate biosynthetic process) — *same* as the seed term; correct.
- **MF (recommended addition): GO:0004733** (pyridoxamine phosphate oxidase activity, EC 1.4.3.5) — a more specific term naming the exact catalytic activity.
- **Granularity relative to the seed:** The process term GO:0042823 is at the **same** (correct) level; the recommended MF term GO:0004733 is **more specific** and sits in a concordant branch (it names the activity that *executes* the process). The two are mutually consistent.

In short, Q88NS5 is a **canonical PNPOx**: it uses FMN as the immediate electron acceptor and O_2 as the ultimate electron acceptor to oxidize pyridoxine 5'-phosphate (PNP) or pyridoxamine 5'-phosphate (PMP) to pyridoxal 5'-phosphate (PLP), the biologically active vitamin-B6 cofactor.

Key Findings

Finding F001 — pdxH (Q88NS5) is a bona fide pyridoxine/pyridoxamine 5'-phosphate oxidase directly in PLP biosynthesis; GO:0042823 is supported.

Subfamily placement is exact. Q88NS5 (215 aa) maps to PANTHER subfamily **PTHR10851:SF0 ("PYRIDOXINE-5'-PHOSPHATE OXIDASE")** — the *same* subfamily as the biochemically and structurally characterized *E. coli* PdxH (P0AFI7). This is the critical control against the "family-vs-subfamily" failure mode. The propagated term was not emitted from a broad ancestral node spanning multiple activities; it was emitted from a tight, single-activity subfamily whose reference member is experimentally characterized. The nearest characterized neighbor is therefore not a distant paralog — it is a direct functional ortholog.

Domain architecture is identical. Both proteins carry the same two-domain organization: Pfam **PF01243 (PNPOx_N)** followed by **PF10590 (PNP_phzG_C)**, with InterPro signatures **IPR000659** / **IPR011576** / **IPR019576** and, critically, the conserved-site signature **IPR019740 (Pyridox_Oxase_CS)**. Matching domain architecture across the full length rules out the "within-superfamily mis-placement" failure mode, in which a query is grafted onto a structurally related but functionally distinct subfamily that shares only part of a fold. Here the

entire architecture — including the family-diagnostic conserved-site motif — is preserved. (The C-terminal PF10590 is also shared with PhzG of phenazine biosynthesis; that potential confound is addressed under *Conflicts* below and is excluded by the PNPOx-specific N-terminal domain, the SF0 call, the gene identity, and the fact that *P. putida* KT2440 is a non-phenazine producer.)

Sequence identity is well within the orthology range. A global alignment of Q88NS5 against *E. coli* PdxH returns **41.5% identity over 212 aligned columns**. For a two-domain FMN-dependent oxidase of ~215 residues, this level of identity across the entire length is comfortably within the range at which molecular function is conserved, and is consistent with the literature observation that members of the PNPOx family share a common three-dimensional fold (PMID: 10903950). A cross-ortholog check against characterized **human PNPO (Q9NVS9)** returned **49.3% identity** — the single highest match among tested references — further reinforcing that the query is a canonical PNPOx.

Active site is fully intact. Of the 25 UniProt-annotated substrate/FMN-binding residues, **20 (80%) are identical** between Q88NS5 and *E. coli* PdxH. The 5 differences are **all conservative** substitutions (E16D, I68V, V69L, Y82F, R88A) that preserve the chemical character of the position. Most importantly, **every critical catalytic and phosphate-binding residue is conserved** (see the Active-Site / Placement Analysis table). The complete conservation of this residue constellation directly rejects the "pseudo-enzyme / loss of activity" failure mode: a protein that had lost catalysis would be expected to show degeneration at precisely these positions; instead all are maintained. UniProt independently assigns **EC 1.4.3.5** to this protein via the curated **HAMAP rule MF_01629**, corroborating the computed active-site conservation with an orthogonal, rule-based annotation.

Process membership is established in the literature. The PNPOx/PdxH reaction is documented as the **terminal step** of de novo PLP biosynthesis in *E. coli* and as part of the B6 salvage pathway (PMID: 12686112; PMID: 10775448). Because Q88NS5 is a direct ortholog of that enzyme, its participation in **GO:0042823 (pyridoxal 5'-phosphate biosynthetic process)** follows directly from its conserved molecular function.

Mechanistic Model / Interpretation

The enzyme catalyzes the last committed reaction of vitamin-B6 (PLP) biosynthesis, oxidizing the 4'-hydroxymethyl group of PNP (or the 4'-aminomethyl group of PMP) to the 4'-aldehyde of PLP, using FMN as the proximal electron acceptor and molecular oxygen as the terminal acceptor, releasing H_2O_2:

```
  Pyridoxine 5'-phosphate (PNP)  ─┐
                                  ├─(FMN, O2)──►  Pyridoxal 5'-phosphate (PLP)  +  H2O2
  Pyridoxamine 5'-phosphate (PMP)─┘
                      ▲
                      │  catalyzed by PdxH / PNPOx (EC 1.4.3.5)
                      │  Q88NS5  ≡  E. coli PdxH (41.5% id, same active site)
```

Position within the pathway:

```
de novo branch:   ... ─► pyridoxine 5'-P (PNP) ──┐
salvage branch:   ... ─► pyridoxamine 5'-P (PMP)─┤
                                                 ▼
                                  [  PdxH / PNPOx  ]  ◄── Q88NS5 (pdxH, P. putida)
                                                 ▼
                                  Pyridoxal 5'-P (PLP)  = active B6 cofactor
                                                 │
                                                 ▼
                        delivered to apo-enzymes (e.g. serine hydroxymethyltransferase)
```

The logical chain from evidence to verdict is:

```
Correct subfamily (PTHR10851:SF0)                ─┐
Identical domain architecture (PF01243+PF10590)  ─┤
41.5% id to characterized E. coli PdxH           ─┼──►  SAME function as characterized ortholog
49.3% id to characterized human PNPO             ─┤      = PNPOx (EC 1.4.3.5, GO:0004733)
All catalytic/FMN/phosphate residues intact      ─┤      = executes GO:0042823
HAMAP MF_01629 → EC 1.4.3.5                       ─┘

                                                          ∴ GO:0042823 SUPPORTED
```

Each of the three TreeGrafter failure modes corresponds to a node this evidence explicitly closes: subfamily placement closes granularity, residue conservation closes pseudo-enzyme, and domain architecture closes mis-placement. The convergence of independent lines —

PANTHER subfamily, Pfam/InterPro architecture, pairwise identity to two characterized orthologs, residue-level conservation, and an orthogonal HAMAP EC assignment — is what elevates the conclusion from "plausible homology" to a confident **SUPPORTED** call.

Evidence Base / Evidence Matrix

#	Citation	Evidence type	Role	Claim tested	Key finding	Organism / context
1	Computational (this analysis)	Structural/ evolutionary + computational	**Supports**	Subfamily placement	Q88NS5 → PANTHER **PTHR10851:SF0**, same subfamily as *E. coli* PdxH; identical PF01243+PF10590 / InterPro architecture incl. IPR019740 conserved site	*P. putida* KT2440 vs *E. coli*
2	Computational (this analysis)	Computational / sequence	**Supports**	Orthology & active-site conservation	41.5% id to *E. coli* PdxH (and 49.3% to human PNPO); 20/25 binding residues identical (5 conservative); all critical catalytic/FMN/ phosphate residues conserved	Pairwise global alignment
3	UniProt Q88NS5 / HAMAP MF_01629	Review/ database (rule-based)	**Supports**	Catalytic activity assignment	Assigns **EC 1.4.3.5** to Q88NS5 via curated HAMAP family rule	Expert-curated rule
4	PMID: 12686112	Structural / direct assay (di Salvo 2003)	**Supports**	Process membership (PLP biosynthesis)	"This reaction serves as the terminal step in the de novo biosynthesis of PLP in E. coli and as a part of the salvage pathway..."	*E. coli*, structure & mechanism

#	Citation	Evidence type	Role	Claim tested	Key finding	Organism / context
5	PMID: 10903950	Structural / evolutionary (Safo 2000)	**Supports**	Fold conservation across family	"A high degree of sequence homology among the 15 known members of the PNPOx family suggests that all members of this group have similar three-dimensional folds."	*E. coli* PNPOx–FMN, 1.8 Å
6	PMID: 10775448	Direct assay (Yang 2000)	**Qualifies / supports**	FMN requirement & PLP targeting	ApoPNPOx (FMN removed) is catalytically inactive; a tight non-catalytic PLP site targets PLP to apo-SHMT in vivo	*E. coli* recombinant
7	PMID: 15858270	Structural (Safo 2005)	**Supports**	Active-site architecture / mechanism	Tetragonal ePNPOx structure resolves substrate binding → catalysis → product release; fold shared with *E. coli* and human enzymes	*E. coli*, 2.6 Å

How the literature supports the verdict. The four primary papers all characterize the *E. coli* (and, by comparison, human) PNPOx enzyme that is the direct ortholog of Q88NS5. PMID: 12686112 and PMID: 15858270 establish the catalytic mechanism and active-site architecture, which define the exact residues scored as conserved in the query. PMID: 10903950 establishes that the PNPOx family shares a common fold, licensing the transfer of function across homologs. PMID: 10775448 establishes the obligate FMN dependence and the physiological role

of delivering PLP to client apo-enzymes. None of these papers challenges the verdict; together they provide the characterized reference against which the query's intact active site is a strong positive signal.

Active-Site / Placement Analysis (computed provenance)

Global Needleman–Wunsch alignment of Q88NS5 (215 aa) vs *E. coli* PdxH P0AFI7 (218 aa): **41.5% identity over 212 ungapped columns**. UniProt-annotated substrate/FMN binding residues (E. coli numbering → P. putida numbering):

E. coli	res	P. putida	res	conserved
R14	R	R9	R	✔
R15	R	R10	R	✔
E16	E	D11	D	conservative
Y17	Y	Y12	Y	✔
R67	R	R64	R	✔
I68	I	V65	V	conservative
V69	V	L66	L	conservative
L70/L71	L/L	L67/L68	L/L	✔
K72	K	K69	K	✔
Y82	Y	F79	F	conservative (aromatic)
T83	T	T80	T	✔
R88	R	A85	A	differs (peripheral)
K89	K	K86	K	✔
Q111	Q	Q108	Q	✔
Y129	Y	Y126	Y	✔
R133	R	R130	R	✔
S137	S	S134	S	✔
Q146/S147	Q/S	Q143/S144	Q/S	✔
W191	W	W188	W	✔
R197	R	R194	R	✔
L198/H199	L/H	L195/H196	L/H	✔
R201	R	R198	R	✔

Result: 20/25 (80%) binding residues identical; all 5 differences conservative. Every critical phosphate/FMN/catalytic residue (R9/R10, Y12, R64, K69, R130, R194, H196, R198) is intact and correctly spaced. The catalytic machinery is present — the protein is a catalytically competent PNPOx, not a pseudo-enzyme.

Cross-ortholog check: Q88NS5 vs characterized **human PNPO (Q9NVS9)** = **49.3% identity**, its single highest match among tested references and higher than to *E. coli* PdxH (41.5%), reinforcing that the query is a canonical PNPOx.

> **Methodological caveat (provenance honesty):** The pairwise global-alignment %-identity metric used here is a simple match/mismatch scheme that can inflate apparent identity for unrelated proteins (under the same scoring, ~39–42% "identity" was also obtained against nitronate monooxygenase and isovaleryl-CoA dehydrogenase). Consequently, the **sibling-vs-PNPOx discrimination rests on the authoritative HMM-based PANTHER subfamily call (PTHR10851:SF0) and the active-site residue conservation, not on raw %-identity.** The %-identity values are reported as supporting, not decisive, evidence. Readers reproducing the per-position mapping should regenerate the Q88NS5–P0AFI7 alignment from the two accessions, as residue-numbering offsets can shift by a few positions.

GO Curation Implications

Lead curation action (requires curator verification): RETAIN — and optionally MAKE-MORE-SPECIFIC by adding a molecular-function term.

- **GO:0042823** (pyridoxal 5'-phosphate biosynthetic process, BP) — **RETAIN.** The IEA propagation is correct. A curator could reasonably upgrade the evidence basis if author-verified or experimental support is cited, but IEA is defensible as-is.
- **Add GO:0004733** (pyridoxamine phosphate oxidase activity, MF; EC 1.4.3.5) — recommended, to capture the specific catalytic activity that executes the retained process. UniProt/HAMAP already assert EC 1.4.3.5 for this entry. This makes the annotation set more informative without contradicting the propagated term.
- This is a **core** function of the protein.

- Do **not** generalize, remove, replace-with-sibling, or treat-as-non-core — none is warranted. Per the brief, "protein binding" is explicitly avoided as a recommendation and is not applicable here.

Limitations and Knowledge Gaps

1. **No *P. putida*-specific biochemical assay.** All direct functional characterization (kinetics, structures, mechanism) comes from the *E. coli* and human enzymes; the *P. putida* ortholog is assigned by homology, not by direct assay. This is a standard and well-justified transfer given 41.5–49.3% identity and complete active-site conservation, but it remains a computational/transferred rather than experimental assignment for this exact protein.
2. **%-identity metric is crude.** As noted above, the pairwise global-alignment identity scoring can inflate values for remote or unrelated proteins; the decisive evidence is the HMM subfamily placement and active-site conservation, not the raw percentages.
3. **Residue-numbering cross-walk.** The Q88NS5 ↔ *E. coli* position mapping depends on the pairwise alignment; small offsets are possible and should be re-verified against a fresh alignment before publishing exact position labels.
4. **Paralog/sibling confusion (PhzG) — low residual risk.** The C-terminal PF10590 is shared with PhzG (phenazine biosynthesis). This is excluded by the PNPOx-specific N-terminal PF01243 domain, the SF0 subfamily call, the gene name *pdxH*, the intact active site, and the fact that *P. putida* KT2440 is a non-phenazine producer (it lacks the *phzABCDEFG* operon). A direct Q88NS5-vs-characterized-PhzG alignment was attempted but the crude pairwise metric is not discriminating for remote homologs; the HMM subfamily placement is the reliable call.
5. **Substrate-preference nuance.** PNPOx oxidizes both PNP and PMP; the relative in vivo contribution of the de novo vs salvage branches in *P. putida* is not established here and is not required for the GO:0042823 call.

Proposed Follow-up Experiments / Actions

Concrete, actionable next steps, in order of efficiency:

1. **AlphaFold structural superposition (fast, purely computational).** Superpose the Q88NS5 AlphaFold model onto an *E. coli* PdxH crystal structure (e.g., PDB 1DNL / 1G76) and confirm the FMN pocket and the arginine phosphate-binding cluster are geometrically congruent. This corroborates the active-site conservation reported above without wet-lab work.
2. **Full PNPOx-family MSA.** Build a multiple-sequence alignment across characterized PNPOx-family members to cement subfamily placement and replace the crude pairwise %-identity metric with a principled tree/HMM-based placement.
3. **Direct enzyme assay (decisive).** Express recombinant Q88NS5 and measure FMN-dependent oxidation of PNP/PMP to PLP (spectrophotometric A_{388} PLP formation, fluorescence, or coupled H_2O_2 detection). This is the single most decisive confirmatory experiment.
4. **Genetic complementation.** Test whether *P. putida pdxH* (Q88NS5) rescues an *E. coli ΔpdxH* B6-auxotroph. A positive result experimentally locks in both molecular function and process membership.

Bottom Line

The TreeGrafter/PANTHER propagation of **GO:0042823** to pdxH (Q88NS5) is **correct**: the protein is a genuine, catalytically intact pyridoxine/pyridoxamine 5'-phosphate oxidase that performs the terminal step of PLP biosynthesis. None of the three propagation failure modes (granularity, pseudo-enzyme, mis-placement) applies. Recommended curation: **retain GO:0042823**, optionally adding the specific molecular-function term **GO:0004733 (EC 1.4.3.5)**.

References

- PMID: 12686112 — *Structure and mechanism of Escherichia coli pyridoxine 5'-phosphate oxidase.*
- PMID: 10903950 — *X-ray structure of Escherichia coli pyridoxine 5'-phosphate oxidase complexed with FMN at 1.8 Å resolution.*
- PMID: 15858270 — *Structure of Escherichia coli pyridoxine 5'-phosphate oxidase in a tetragonal crystal form: insights into the mechanistic pathway of the enzyme.*

- PMID: 10775448 — *Tight binding of pyridoxal 5'-phosphate to recombinant Escherichia coli pyridoxine 5'-phosphate oxidase.*
